Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/26/2007
PI Name:	Kailasanath, Kazhikathra Ph.D.		
Project Title:	Fire Suppression And Safety In Reduced Gravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20375	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-B
Start Date:	10/01/2003	**End Date:**	03/31/2007
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: changed end date to 3/31/2007 per PI (1/07) NOTE: Added to Task Book 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Abbud Madrid, Angel (University of Colorado Boulder)		
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Task Description:	Conducting space-based testing of fire suppression in large compartments is a daunting task from both the safety and the cost points of view. Simulations using a validated and verified numerical tool will provide valuable information that can complement and reduce the number of tests needed as well as help design better and safer experiments for testing fire-safety policies and practices. The primary objective of this research effort is to develop and demonstrate a computational tool for predicting the suppression of small or incipient fires in large (rack-scale) compartments. A secondary objective was to use this tool to understand and quantify the relative suppression effectiveness of CO2 and water-mist on fires in different gravitational environments. Over the years, a unique capability has been developed at the Naval Research Laboratory to simulate both the detailed physics and chemistry of flames and fires as well as the flows in and around complex geometries such as those inside a space habitat. Previously, we have also used our models to study fire dynamics and suppression in a surface ship compartment. The approach we have adopted in this project is to modify, adapt and develop further the numerical simulation tool originally developed for compartment fire and shipboard smoke transport simulations. The original simulation tool included only a simple heat release model, simplified transport and other physical models and did not allow for any local mesh refinement. These are areas of improvement needed before use of the tool effectively for predicting fire suppression in normal and reduced gravity environments. The droplet and particulate transport sub-models and solution approaches also needed further verification and validation. Aspects of the model and simulation results for specific conditions were validated using experiments conducted at and by the Center for Commercial Applications of Combustion in Space (CCACS) of the Colorado School of Mines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has both basic scientific value and direct application to NASA goals to develop "effective methods of extinguishments of fires with minimal residual damage" and "maximize fire safety on orbiting vehicles." It specifically addresses several issues identified during a NASA Workshop on Spacecraft Fire Safety. These include: characterizing the flow field in spacecraft compartments, investigating the transport of fire-fighting agents and their interaction with the air and fuel surrounding a fire. This research also provides additional basic information on fire suppression and thus enhances our knowledge base for fighting fires on Earth and in Spacecraft and extraterrestrial environments. Thus, it also addresses one of the goals of NASA: to "enhance the knowledge base that contributes to earth-based technologies".
Task Progress:	During this time period, we have made further developments of NRL's Fire and Blast Mitigation code, FBM, conducted numerical simulations, compared the results of the simulations to experimental data from CCACS. Further developments to the code include better transport and chemistry models; inclusion of a soot and radiation model and implementation of an Eulerian sectional approach to simulate the transport of water-mist droplets. A simple and cost-effective approach for calculating transport properties is to use the Sutherland's law for viscosity and derive the other diffusive transport properties using constant Prandtl and Schmidt numbers. However, in our preliminary simulations and verification studies, this resulted in large changes in the maximum temperature for small changes in these numbers. Therefore, a more detailed transport module has been incorporated into the code. In this approach, we calculate the transport properties for individual species and then use this information to compute the mixture transport properties for viscosity, thermal conductivity, and diffusion coefficients. We have retained the simpler approach as an option, just in case we can calibrate it with experimental data and use it when approximate but quick turn-around answers are needed. The simpler flame sheet approximation used initially for the chemistry has been replaced with a finite rate two-step model. The chemistry integrator has also been upgraded with that used previously in our NRL FLAME3D model. In this approach, the actual species production and destruction rates are calculated on a local basis and will allow us to incorporate other multi-step chemistry models, if available for simulation of specific fire scenarios of interest to NASA. A simple soot model (including soot radiation) has been included. This model solves two transport equations for the number density of soot particles and the volume fraction of soot. Source terms representing soot nucleation, coagulation, surface growth, and oxidation then complete the model. In addition, an optically thin radiation model was also incorporated, based on CO2, H2O and Soot radiation. The thermodynamics model has been changed from constant ratio of specific heats, to temperature-dependent curve-fits for the individual species enthalpies. The code now uses the thermodynamic curve-fits that are set up using the Chemkin translator. The geometry modeled in the simulations was that of a small co-flowing burner within the larger CCACS enclosure with methane flowing through an inner tube and air flowing through an outer tube. The focus of the first set of simulations was to establish the level of detailed physics required in the simulations to achieve satisfactory comparisons with the experiments. The result of this study showed that the two-reaction global model and the more complex mixture-dependent properties are both vital in predicting a reasonable flame shape and unsteadiness. Thermal properties also contribute to a lesser extent to the flame shape and dynamics. In particular, the onset of unsteadiness is intricately linked to all three models. Then experiments and simulations were carried out for a range of flow conditions. The flame height was found to grow linearly with flow velocity up to 4 cm/s, after which the flames became unsteady and tended to flicker. Although the exact location of the visual flame height is hard to determine in the simulations and compare, the trend in the simulations follows the experimental trend very closely. The final set of simulations was focused on the suppression of these jet diffusion flames using water-mist. Both a steady flame as well as a pulsating flame were studied. Water-mist was injected off-axis from above the burner. In addition to the effect of vaporization and heat extraction, the downwash created by the mist injection has a dramatic effect on the flame. Even the nominally steady (without water-mist injection) flame stretches away from the mist and becomes partially detached from the burner lip. In summary, a fully three-dimensional, unsteady, multiphase simulation tool has been developed for the studying the suppression of small flames in a large, complex enclosure with obstructions. The challenge has been to retain a sufficient level of detail to accurately model the small fire and still keep the problem of simulating the fire in a large enclosure computationally tractable. Additional studies are needed to characterize the relative effectiveness of various fire suppressants injected from arbitrary locations within the enclosure.

Bibliography Type:	Description: (Last Updated: 01/29/2007)
Papers from Meeting Proceedings	Schwer D, Kailasanath K, Abbud-Madrid A. "Development of a fire suppression simulation tool for various low-gravity environments." 45th AIAA Aerospace Sciences Meeting, Reno, NV, Jan 8-11, 2007. AIAA Paper 2007-0735, Reston , VA : AIAA, 2007. , Jan-2007
Papers from Meeting Proceedings	Schwer D, Kailasanath K, Abbud-Madrid A. "Development of a water-mist fire suppression tool for low-gravity fire suppression modeling." Fall technical Meeting of the Eastern States Section of the Combustion Institute, Orlando, FL, Nov. 13-16, 2005. Proc. Eastern States Section of the Combustion Institute, 2005. , Nov-2005